



Stempedia

The Stempedia is a STEM starter kit that will have your kids coding and building robots in no time. <https://dgit.in/jul20-07>



Taco Robobricks

Robotix's Taco Robobricks are modular toys that also integrate with third party bricks. <https://dgit.in/jul20-08>



Technavio's latest market research report for the STEM toys market

The term STEM also added an A at some point in the acronym to include 'Arts', making it STEAM, since creativity and artistic vision is an important part of the STE(A)M educational approach. We will be referring to STEM as STEAM here on forth.

Some popular STEAM toys from established companies in the space include Fisher-Price's Code-a-Pillar, a caterpillar-styled toy which taught children the basics of programming by adding body links to create a sequence of movements (released in 2016), and Mindstorms by Lego, which was a line of engineering and robotics toys that debuted way back in 1998, well before the STEM/STEAM trend, however, it did adopt the acronym quickly in the future. A few others include the Crayola Color Chemistry Set, Valtech Magna-Tiles, Sphero Specdrums, Particula GoCube, LEGO Voltron, Kano Star Wars The Force Coding Kit, DJI Robomaster S1, and many more. These toys essentially catapulted the industry into the globally present one it is today. Startups from other countries also quickly realised the potential of this market and began creating unique toys to break into the thriving toy segment.

STEAM TOY STARTUPS THRIVE IN INDIA

Sensing the growing potential, several Indian startups have entered the competitive and lucrative STEAM toys market with innovative offer-

ings. Perhaps, the most successful of all is Delhi-based startup, Smartivity Labs which was launched in 2015 by Tushar Amin, Apoorv Gupta, Ashwini Kumar and Rajat Jain. Wanting to create a wholesome, immersive and educational play experience for children, Smartivity Labs was born, specialising in creating STEM educational DIY toys, AR-enabled activities, and IoT toys. They wanted to build products that weren't simply art and craft projects but also offered an understanding of STEAM concepts to children. The company offers learning activity boxes for children between the ages of 4 and 12, designed around scientific principles mixed with real-world applications. They also offer AR-enabled colouring and jigsaw puzzle activities. When the colouring is complete or the puzzle is solved, children can scan the



One of Smartivity Labs' DIY construction kit

result using the Smartivity EDGE app and see an interactive AR avatar of their finished result, complete with the colours they chose to fill in.

Smartivity Labs exports its multi-faceted STEAM toys to over 24 countries including the USA, Canada, Australia, Russia, and others. Since its inception, Smartivity Labs has shipped over 1.5 million products and have also raised a total of \$3 million through three rounds of funding. The company also offers a STEAM Ninja program where children can collaborate with the Smartivity team and help them test future prototypes of toys.

Another wildly successful ed-tech startup creating STEAM toys is Bengaluru-based PlayShifu. The company makes interactive AR toys around STEAM principles and have recorded a baffling 8x revenue growth year-on-year as of 2020. Additionally, the startup has a presence all over the world with over 600 clients around the world, and 400 of them being schools in the US. Founded by Dinesh Advani and Vivek Goyal in 2016, PlayShifu products are designed for students between the ages of 2-12 years, to help them inculcate STEAM skills via fun, interactive AR toys.

Its flagship product is a smart world globe named Orboot which can be utilised by young adults and children to explore the geography and culture of cities or even countries. PlayShifu's portfolio also includes an array of AR puzzles and quest kits named Shifu Plugo. There are many more interactive toys including board games, flashcard games and more in PlayShifu's extensive arsenal of products. The company has a user base of about 2,50,000 children across 15 countries. In a span of merely three years, PlayShifu has become a multi-million dollar company with an 8x year-on-year growth rate. Vivek Goyal, PlayShifu's co-founder told inc42 that they will also be focusing on adding a variety of products that focus on children above 13 years of age to inculcate programming skills in children.